

Formula 1

A Mathematical Analysis to Identify the Top Formula 1 Driver in History

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1. Introduction

When posed the question, “Who is or was the best Formula 1 driver of all time?” an AI agent will likely first mention **Lewis Hamilton**, followed by **Michael Schumacher**, based primarily on their respective world championship titles. If asked for a list of top drivers, it might also include **Juan Manuel Fangio**, **Alain Prost**, **Max Verstappen**, and **Sebastian Vettel**, arranged alphabetically. **Fangio** stands out with five championships, while the others each have four.

Interestingly, many fans regard **Ayrton Senna**, with his three titles, as a superior driver compared to **Alain Prost**, despite the latter’s four championships. This raises the question: how significant are titles in evaluating a driver’s greatness? Should we consider alternative criteria for determining who truly excels?

If we delve deeper, **Juan Manuel Fangio** won five titles in the 1950s, during the formative years of the Formula 1 World Championship. His legacy is so profound that his name has become synonymous with exceptional driving. In French-speaking regions, the phrase “driving like Fangio” denotes flawless driving. A notable reference appears in *The Adventures of Tintin: The Calculus Affair* [1], where a character remarks, “This must be Fangio at the wheel” as a car skillfully avoids a net, reflecting Fangio’s prowess during the height of his career. (In the English edition, it is translated as “They must have a Jack Brabham at the wheel!”)



Figure 1: “This must be Fangio at the wheel!”

Complicating matters further, the rules of Formula 1 have evolved significantly over the years. The number of races has increased (from seven during Fangio’s era to 22 today), and the points system has undergone various changes. For instance, the winner received 8 points until **1960**, then 9 points until **1990**, followed by 10 points until **2009**, and has been set at 25 points since **2010**. Additionally, until **1990**, only the best results counted towards the championship, which allowed Senna to secure the title in **1990** against Prost, despite the latter having more points overall; under modern rules, Prost would have been awarded the title instead.

Ultimately, it is evident that even the most skilled drivers depend significantly on their cars’ performance. Both Alain Prost in **1991** and Lewis Hamilton in **2025** encountered substantial difficulties while driving for Ferrari, so much so that Prost was dismissed from his team after comparing his car to a “truck.” This raises an important question: to what extent is a world title attributable to the car, and how much is a reflection of the driver’s ability?

2. Dataset

2.1 Source

The Formula 1 data, including information on drivers, teams, qualifications, and results, is readily available online, such as on Wikipedia [2]. However, for research purposes, a well-maintained dataset may be more valuable. A suitable dataset can be found on Kaggle [3], although accessing it requires an account that some users may not possess. A publicly accessible version of this dataset is also available on GitHub Training Insights, but to guarantee its ongoing accessibility, I have uploaded a copy to my GitHub repository [4]. This redundancy will ensure that the dataset remains available even if the original site removes it.

2.2 Files

The dataset is organized as nineteen different files : 18 comma-separated value files and a json file. We will only use the 9 files below :

Dataset Files	
	circuits.csv
	constructor_results.csv
	constructor_standings.csv
	constructors.csv
	driver_standings.csv
	drivers.csv
	races.csv
	results.csv
	status.csv

The other files listed below aren't relevant for this study :

Dataset Files	
	fatal_accidents_drivers.csv
	fatal_accidents_marshalls.csv
	lap_times.csv
	pit_stops.csv
	qualifying.csv
	red_flags.csv
	safety_cars.csv
	seasons.csv
	sprint_results.csv
	virtual_safety_car_estimates.json

2.3 Tables

2.3.1 circuits.csv

This file contains information about each circuit where a Formula 1 race has been held.

Column	Description
circuitId	Unique identifier for each circuit
circuitRef	A short reference name for the circuit
name	The official name of the circuit
location	The city where the circuit is located
country	The country where the circuit is located
lat	Latitude of the circuit
lng	Longitude of the circuit
alt	Altitude of the circuit in meters
url	URL for the circuit's Wikipedia page

2.3.2 constructor_results.csv

This file contains the results for each constructor in each race.

Column	Description
constructorResultsId	Unique identifier for each constructor result
raceId	Foreign key to <code>aces.csv</code>
constructorId	Foreign key to <code>constructors.csv</code>
points	Points scored by the constructor in the race
status	Status of the constructor in the race

2.3.3 constructor_standings.csv

This file contains the constructor standings after each race.

Column	Description
constructorStandingsId	Unique identifier for each constructor standing
raceId	Foreign key to <code>aces.csv</code>
constructorId	Foreign key to <code>constructors.csv</code>
points	Total points for the constructor after the race
position	Constructor's position in the standings
positionText	Text representation of the constructor's position
wins	Number of wins for the constructor in the season

2.3.4 constructors.csv

This file contains information about each constructor.

Column	Description
constructorId	Unique identifier for each constructor
constructorRef name	A short reference name for the constructor
nationality	The full name of the constructor
url	The nationality of the constructor
	URL for the constructor's Wikipedia page

2.3.5 driver_standings.csv

This file contains the driver standings after each race.

Column	Description
driverStandingsId	Unique identifier for each driver standing
raceId	Foreign key to races.csv
driverId	Foreign key to drivers.csv
points	Total points for the driver after the race
position	Driver's position in the standings
positionText	Text representation of the driver's position
wins	Number of wins for the driver in the season

2.3.6 drivers.csv

This file contains information about each driver.

Column	Description
driverId	Unique identifier for each driver
driverRef	A short reference name for the driver
number	The driver's car number
code	A three-letter code for the driver
forename	The driver's first name
surname	The driver's last name
dob	The driver's date of birth
nationality	The driver's nationality
url	URL for the driver's Wikipedia page

2.3.7 races.csv

This file contains information about each race.

Column	Description
raceId	Unique identifier for each race
year	The year of the race
round	The round number of the race in the season
circuitId	Foreign key to <code>circuits.csv</code>
name	The name of the Grand Prix
date	The date of the race
time	The time of the race
url	URL for the race's Wikipedia page
fp1_date	Date of Free Practice 1
fp1_time	Time of Free Practice 1
fp2_date	Date of Free Practice 2
fp2_time	Time of Free Practice 2
fp3_date	Date of Free Practice 3
fp3_time	Time of Free Practice 3
quali_date	Date of Qualifying
quali_time	Time of Qualifying
sprint_date	Date of Sprint Race
sprint_time	Time of Sprint Race

2.3.8 results.csv

This file contains the results for each driver in each race.

Column	Description
resultId	Unique identifier for each result
raceId	Foreign key to <code>races.csv</code>
driverId	Foreign key to <code>drivers.csv</code>
constructorId	Foreign key to <code>constructors.csv</code>
number	The driver's car number
grid	The driver's starting grid position
position	The driver's finishing position
positionText	Text representation of the driver's finishing position
positionOrder	The driver's finishing position in order
points	Points scored by the driver in the race
laps	Number of laps completed by the driver
time	The driver's total race time
milliseconds	The driver's total race time in milliseconds
fastestLap	The lap number of the driver's fastest lap
rank	The rank of the driver's fastest lap
fastestLapTime	The driver's fastest lap time
fastestLapSpeed	The speed of the driver's fastest lap
statusId	Foreign key to <code>status.csv</code>

2.3.9 status.csv

This file contains information about the finishing status of a driver in a race.

Column	Description
statusId	Unique identifier for each status
status	A description of the finishing status

2.4 Entity Relationship Diagram Model

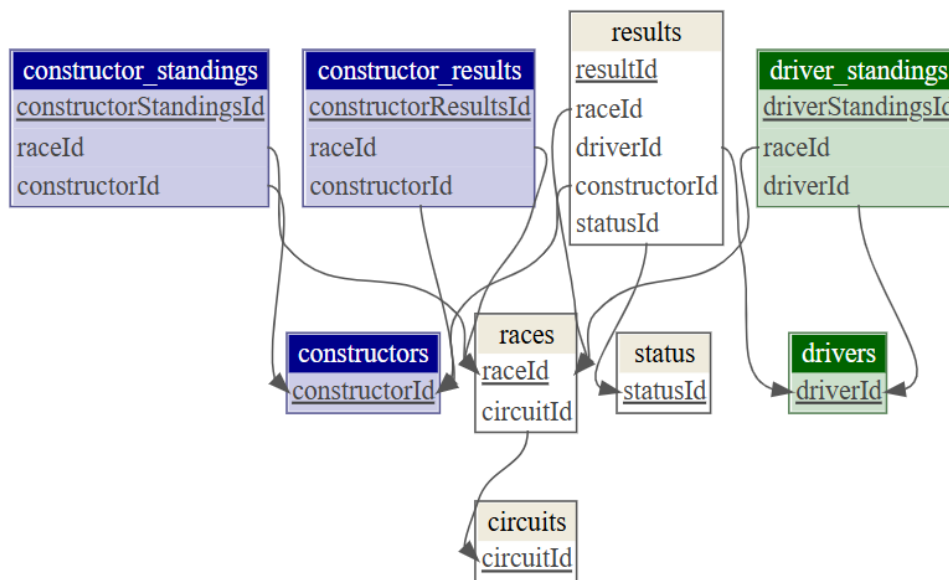


Figure 2: Entity Relationship Diagram Model

2.5 Data Issues

The following figure show that 14 rows in the **driver_standing** table have a position different to the positionText, with only one being “D” (which seems to be for Disqualified) :

driverStandingsId	raceId	driverId	points	position	positionText	wins
11031	223	30	78	26	D	5
73285	1145	859	0	\N	-	0
73286	1145	864	0	\N	-	0
73287	1145	4	0	\N	-	0
73288	1145	832	0	\N	-	0
73289	1145	862	0	\N	-	0
73290	1145	865	0	\N	-	0
73310	1146	4	0	\N	-	0
73785	1169	840	0	\N	-	0
73786	1169	4	0	\N	-	0
73787	1169	822	0	\N	-	0
73788	1169	865	0	\N	-	0
73789	1169	857	0	\N	-	0
73790	1169	807	0	\N	-	0

The raceId 223 and driverId 30 correspond to Michael Schumacher who was disqualified after the Grand Prix of Europe in 1997 :

raceId	year	name	driverId	forename	surname
223	1997	European Grand Prix	30	Michael	Schumacher

However this wasn’t the only disqualification from the history of Formula 1. The table below shows 5 rows from the results table containing 151 observations related to a disqualification (statusId of 2) :

resultId	raceId	driverId	constructorId	grid	position	positionText	positionOrder	points	laps	statusId
22	18	22	11	10	\N	D	22	0	58	2
499	41	13	6	5	\N	D	21	0	51	2
500	41	21	4	9	\N	D	22	0	51	2
1005	64	27	13	21	\N	D	21	0	66	2
1006	64	33	13	18	\N	D	22	0	66	2
1028	65	9	2	9	\N	D	22	0	69	2

More interestingly, the raceId 223 and driverId 30 from the results dataset don’t indicate a disqualification but a collision :

statusId	status
4	Collision

Now if we look at fatal accidents (statusId = 104), we can only find 3 results :

resultId	grid	positionOrder	laps	statusId	year	name	forename	surname	status
11303	23	25	0	104	1982	Canadian Grand Prix	Riccardo	Paletti	Fatal accident
14953	23	23	9	104	1974	United States Grand Prix	Helmuth	Koinigg	Fatal accident
16133	0	23	0	104	1970	Italian Grand Prix	Jochen	Rindt	Fatal accident

We know that more drivers died during a Formula 1 Grand Prix.

For example Ayrton Senna's status in the 1994 San Marin GP was marked as accident (statusId = 3) :

resultId	grid	positionOrder	laps	statusId	year	name	forename	surname	status
5216	1	22	5	3	1994	San Marino Grand Prix	Ayrton	Senna	Accident

Which can indicate that the fatal accidents only report drivers who died on-site during a race or qualifications.

The following table shows a list of statuses, only the first 12 are displayed :

statusId	status
1	Finished
2	Disqualified
3	Accident
4	Collision
5	Engine
6	Gearbox
7	Transmission
8	Clutch
9	Hydraulics
10	Electrical
11	+1 Lap
12	+2 Laps

We will only want to have the following statuses : "Finished", "Abandoned", "Lapsed" and "Disqualified". The "Lapsed" meaning the driver finished the race but with 1 or more laps behind the winner. The "Abandoned" includes all sort of mechanical problems and collisions, accidents... The "Disqualified" status is the same as "Abandoned" but the driver then didn't score any point.

The only exception found is Stirling Moss during the 1959 French GP who scored one point (best lap). He was disqualified during the race and kept his only point for the best lap :

resultId	points	year	name	forename	surname	status
18372	1	1959	French Grand Prix	Stirling	Moss	Disqualified

3. Data Transformation

We will now preprocess the Formula 1 dataset, focusing on optimizing the data frames for analysis by removing unnecessary columns, joining relevant data, renaming columns for clarity, and calculating driver age.

3.1 Removing Unnecessary Columns

To streamline the dataset, unnecessary columns from various data frames are removed. The goal is to retain only relevant information that will contribute to the overall analysis.

Circuits: The columns `lat`, `lng`, and `url` are omitted from the circuits data frame as they do not provide essential information for the analysis of race results.

Constructor Results & Standings: In the `constructor_results` and `constructor_standings` data frames, the `status` and `positionText` columns are excluded, respectively, as they are not required for further analysis.

Drivers' Columns: Columns related to drivers, such as `number` and `url` in the drivers data frame, are removed from `driver_standings`, simplifying the dataset.

Races Table: The races data frame is streamlined to include only the columns `raceId`, `year`, `round`, `circuitId`, `name`, and `date`, ensuring relevant race information is retained while discarding less useful details.

Results' Columns: A significant reduction in the results data frame is performed by dropping columns such as `number`, `time`, `milliseconds`, as well as fastest lap information (`fastestLap`, `fastestLapTime`, `fastestLapSpeed`), which may not be needed for the analysis of race outcomes.

3.2 Modifying statuses

As noted in the previous section, the status table includes various statuses that may not be relevant to this study. We will update these to reflect more accurate values.

3.3 Joining Tables

To create a comprehensive Formula 1 dataset, the results data frame is merged with several other data frames using left joins based on common identifiers.

The results data frame is join together with races, drivers, constructors, and status, linking them through `raceId`, `driverId`, `constructorId`, and `statusId`, respectively. This process allows for the integration of race and participant information into a single dataset.

3.4 Finalizing the Dataset

After merging the data frames, additional columns that are no longer necessary (`raceId`, `driverId`, `constructorId`, `circuitId`) are excluded to reduce clutter in the dataset.

3.5 Renaming Columns

To enhance clarity and facilitate a better understanding of the dataset, several columns are renamed. The following renaming conventions are employed:

`name.x` becomes `circuitName` `name.y` becomes `constructorName` `nationality.x` becomes `driverNationality` `nationality.y` becomes `constructorNationality`

These changes improve readability and usability of the dataset for analysis.

3.6 Calculating Driver Age

Driver age is computed by converting the date and dob columns into Date format. The age is calculated using the difference in weeks between the two dates. The result is then divided by 52.25 to convert weeks into years, and the floor() function is applied to round down to the nearest integer. This new column, driverAge, is added to the dataset.

3.7 Concatening Driver Names

The driver's full name is created by concatenating the forename and surname. The original two columns are then removed from the dataset.

3.8 Cleaning Up Date Columns

Finally, the date and dob columns are removed from the dataset, as they are no longer necessary after calculating driver age.

Overall, these transformations enhance the dataset's efficiency and layout, preparing it for insightful analysis in subsequent sections of the report.

3.9 Formula 1 Dataset

The dataset displayed below is a snippet of the final transformed dataset, showcasing only 8 out of the 16 columns for enhanced clarity:

resultId	positionOrder	points	year	circuit	driverName	driverAge	constructorName
1	1	10	2008	Australian Grand Prix	Lewis Hamilton	23	McLaren
2	2	8	2008	Australian Grand Prix	Nick Heidfeld	30	BMW Sauber
3	3	6	2008	Australian Grand Prix	Nico Rosberg	22	Williams
4	4	5	2008	Australian Grand Prix	Fernando Alonso	26	Renault
5	5	4	2008	Australian Grand Prix	Heikki Kovalainen	26	McLaren
6	6	3	2008	Australian Grand Prix	Kazuki Nakajima	23	Williams

The formula1 dataset contains the following columns :

Column	Description
resultId	Unique identifier for each result
grid	The driver's starting grid position
positionOrder	The driver's finishing position in order
points	Points scored by the driver in the race
laps	Number of laps completed by the driver
rank	The rank of the driver's fastest lap
statusId	Foreign key to <code>status.csv</code>
year	The year of the race
round	The round number of the race in the season
circuit	The official name of the circuit
driverNationality	The driver's nationality
constructorName	The full name of the constructor
constructorNationality	The nationality of the constructor
status	A description of the finishing status
driverAge	The driver's age at the time of the race
driverName	The driver's full name

4. Exploratory Data Analysis

4.0 Grands Prix

The formula 1 world championship started in 1950, and has been held since then with no interruption. As of 2026, 76 seasons has been held, with 1,149 races where 864 drivers were involved. Only 115 won a race, and 35 won the world champion title :

Number of Seasons	Number of Races	Number of Drivers	Number of Winners	Number of Champions
76	1,149	864	115	35

Out of the 35 world champions, 17 of them won the title several times. The plots below show the number of drivers per title or win category :

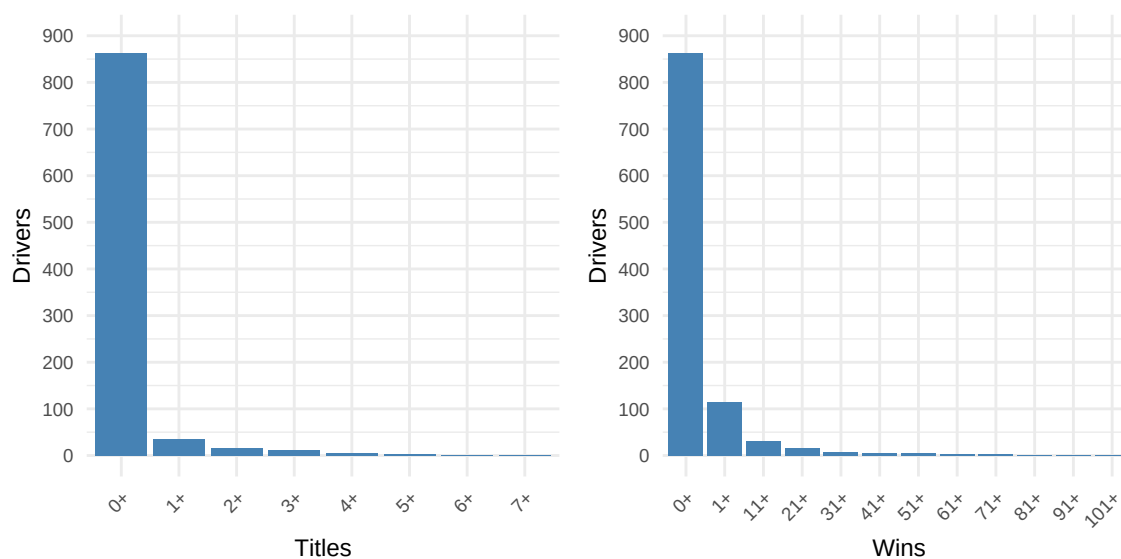


Figure 3: Number of Titles or Wins per Driver

If we remove the drivers who didn't win any title or Grand Prix, we can see the distinction in the plots below :

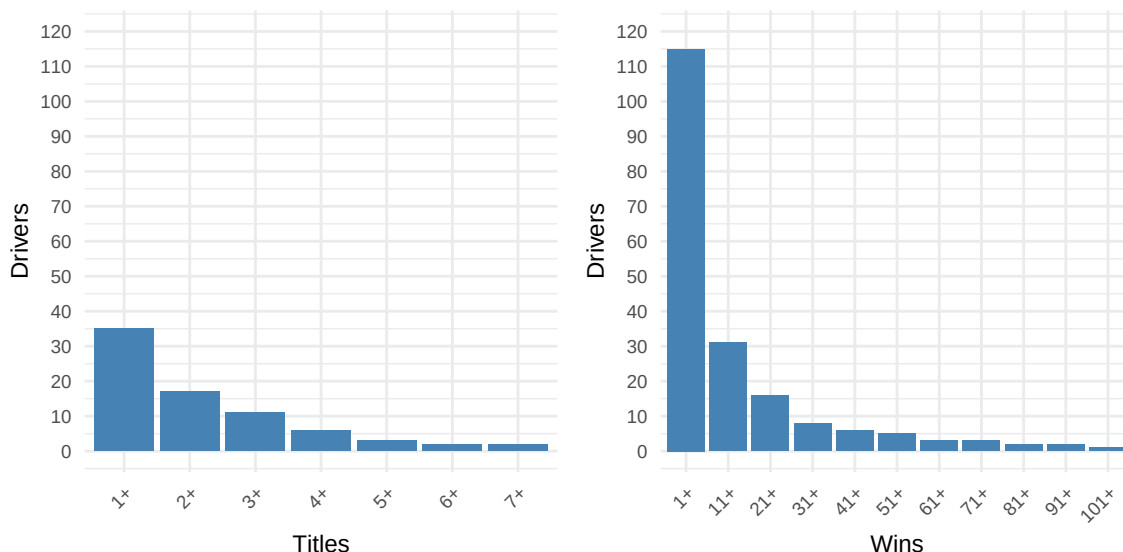


Figure 4: Number of Titles or Wins per Driver (who won at least one race or title)

The plots above show that most drivers didn't win any title. For those who won at least one, half won only one title while the other won 2 or more titles.

Very few drivers were able to become world champion while winning less than 11 Grand prix in their whole career, and with no surprise they were all competing in the early years, the most recent title being won by Keke Rosberg in 1982, Rosberg who won 5 Grand Prix in his whole career :

driverName	titles	wins	year
 Nino Farina	1	5	1950
 Mike Hawthorn	1	3	1958
 Phil Hill	1	3	1961
 John Surtees	1	6	1964
 Denny Hulme	1	8	1967
 Jochen Rindt	1	6	1970
 James Hunt	1	10	1976
 Jody Scheckter	1	10	1979
 Keke Rosberg	1	5	1982

On the other side, we also have drivers who, while winning many races (more than 10), didn't win any title. The list below shows that, unlike for the low-wins world champions, some of them were competing in the early years of the F1. Notably, Carlos Reutemann and above all, Stirling Moss won 16 Grand Prix over a career of 11 years starting in 1951 and ending in 1961. As a comparison the champion Fangio won 24 races and 5 titles at the same period (1950-1958) :

driverName	wins	careerstart	careerend	careerlength
 Felipe Massa	11	2002	2017	15
 David Coulthard	13	1994	2008	15
 Rubens Barrichello	11	1993	2011	19
 Carlos Reutemann	12	1972	1982	11
 Stirling Moss	16	1951	1961	11

4.0 Winning Probability

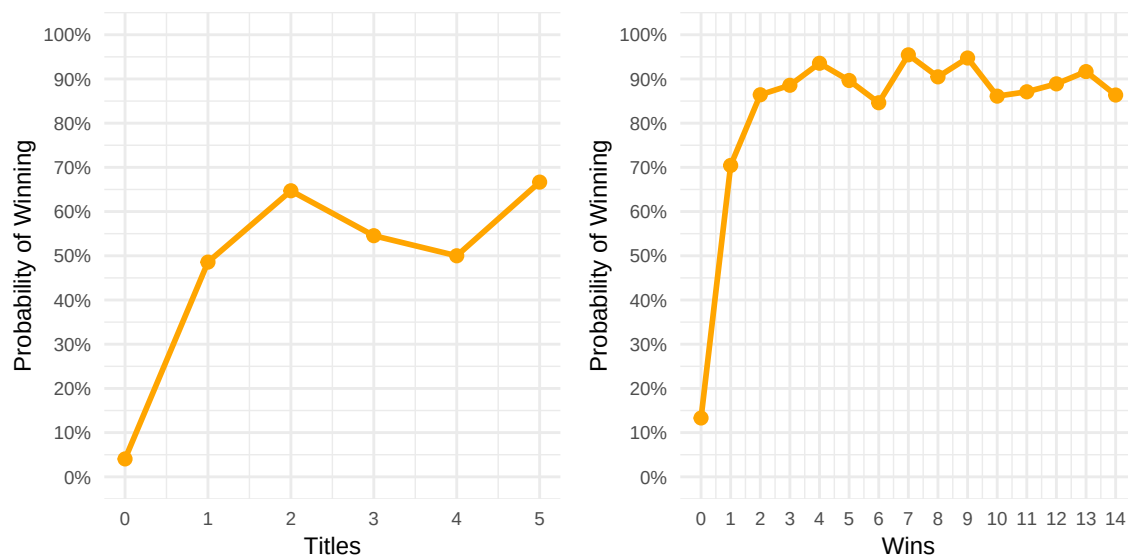


Figure 5: Probability of Winning

The plots above show that the probability of winning is very low for drivers who didn't win a title, and stays about 50% for the other ones. So each time a driver wins a title, he has 50% of chances to win another one. The 100% probability at 6 titles is not very accurate given the low number of drivers who won more than 5 titles (only 3 including Fangio).

4.0 Constructors

The table below shows the number of constructors involved in the championship over the years ;

Number of Constructors	Points	Wins	Titles
211	106	47	18

As of 2025, 211 constructors participated in the world championships. 106 of them won at least one point, 47 won a race and 18 won a title.

And this is a list of the constructors who won the championship at least once :

constructorName	titles	wins	points	start	end	seasons
 Ferrari	22	249	11,451.27	1950	2025	76
 McLaren	11	199	7,797.50	1968	2025	56
 Mercedes	9	131	8,154.64	1954	2025	18
 Williams	9	114	3,765.00	1975	2025	50
 Red Bull	6	130	8,083.00	2005	2025	21
 BRM	3	17	537.50	1951	1977	23
 Renault	2	35	1,777.00	1977	2020	24
 Tyrrell	2	23	711.00	1970	1998	29
 Cooper-Climax	2	12	336.50	1959	1968	9
 Brabham-Repco	2	8	175.00	1966	1969	4
 Team Lotus	1	45	995.00	1958	1994	29
 Benetton	1	27	861.50	1986	2001	16
 Lotus-Climax	1	22	281.00	1961	1967	7
 Alfa Romeo	1	11	361.00	1950	2023	16
 Lotus-Ford	1	11	209.00	1963	1971	6
 Maserati	1	9	313.14	1950	1960	11
 Matra-Ford	1	9	130.00	1968	1969	2
 Brawn	1	8	172.00	2009	2009	1

Ferrari stands out prominently with 22 titles and 249 wins, indicating a dominant presence in the sport since 1950. McLaren and Mercedes follow, with 11 and 9 titles respectively. Constructors like Williams and Red Bull have also made significant contributions, evidenced by their multiple wins and points. However, it's important to note that constructors such as Mercedes and Alfa Romeo did not participate consistently throughout the years. For instance, Mercedes withdrew from racing after the 24h of Le Mans disaster in 1954 and returned to the tracks only in 2010. Similarly, Alfa Romeo has had gaps in its participation, despite its historical significance, which reflects the varying levels of engagement among different teams over the decades.

Now the constructors results over time. Comparison between constructors who won most points from first to second :

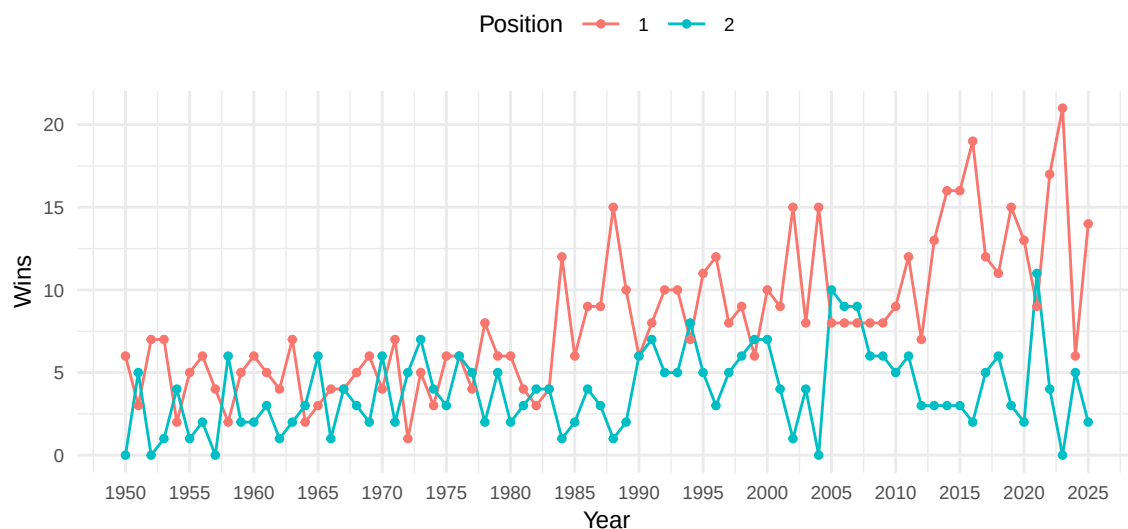


Figure 6: Constructor Results

And the driver results as a comparison :

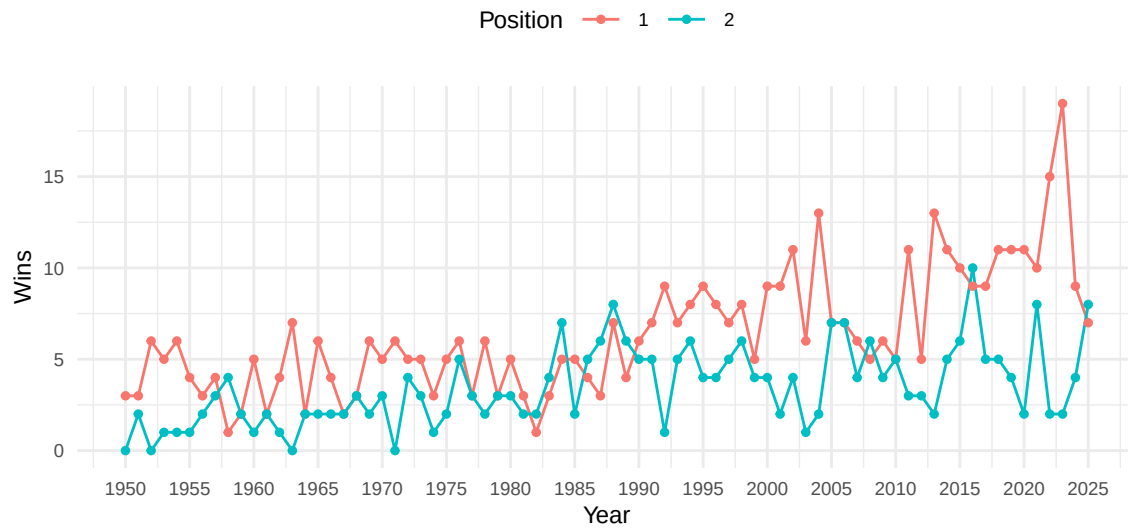


Figure 7: Driver Results

4.1 Top Drivers

4.1.1 Names

If we define the leading drivers as those with the highest number of titles, and among those who are world champions with an equal number of titles, we can use the number of wins to differentiate them. This results in the following table displaying the 17 drivers who won at least 2 titles (as of 2025) :

driverName	titles	wins	races	seasons	points	titlespc	winspc
 Lewis Hamilton	7	105	380	19	4,955.50	0.368	0.276
 Michael Schumacher	7	91	308	19	1,566.00	0.368	0.295
 Juan Fangio	5	24	58	8	279.14	0.625	0.414
 Max Verstappen	4	71	233	11	3,301.50	0.364	0.305
 Sebastian Vettel	4	53	300	16	3,098.00	0.250	0.177
 Alain Prost	4	51	202	13	798.50	0.308	0.252
 Ayrton Senna	3	41	162	11	614.00	0.273	0.253
 Jackie Stewart	3	27	100	9	360.00	0.333	0.270
 Niki Lauda	3	25	174	13	420.50	0.231	0.144
 Nelson Piquet	3	23	207	14	485.50	0.214	0.111
 Jack Brabham	3	14	129	16	261.00	0.188	0.109
 Fernando Alonso	2	32	428	22	2,380.00	0.091	0.075
 Jim Clark	2	25	73	9	274.00	0.222	0.342
 Mika Häkkinen	2	20	165	11	420.00	0.182	0.121
 Emerson Fittipaldi	2	14	149	11	281.00	0.182	0.094
 Graham Hill	2	14	179	18	289.00	0.111	0.078
 Alberto Ascari	2	13	36	6	140.00	0.333	0.361

For each driver, we observe the number of titles, total races won, total races competed, the number of seasons or years they participated, the total points earned (only those that count toward the championship), the ratio of titles to seasons, and the ratio of wins to races.

Although the number of titles and races won clearly shows that Michael Schumacher and Lewis Hamilton outshine other drivers, the same is not necessarily the case for points. When we rank drivers by points, both Verstappen and Vettel surpass Schumacher. Even Alonso who won 2 titles, compared with the 7 by Schumacher, has more points than the latter.

To add another layer of complexity, if we look at the ratio of titles to seasons or the ratio of wins to races, Juan Manuel Fangio emerges as the top performer. Even Ascari, who appears at the bottom of the list, has the second best ratio win to races.

Now if we look at the first 9 pilots, only 2 of them (Fangio and Stewart) were racing at the earliest days of Formula 1. On the opposite, if we look at the last 8 drivers, 4 of them were driving at the earliest days : Brabham, Clark, Hill and Ascari. We can also notice that most of these early drivers had a career under 10 years, when modern days drivers have a career that last more than 10 years if not 20. Only notable exception is Graham Hill who had a long career of 18 years from 1958 to 1975.

We will now see the careers and titles of these 17 drivers across the years.

4.1.2 Timeline

The plot below illustrates the duration and timeline of careers without indicating any breaks; for instance, Alain Prost did not compete in 1992, and Schumacher was absent from the tracks from 2007 to 2009.

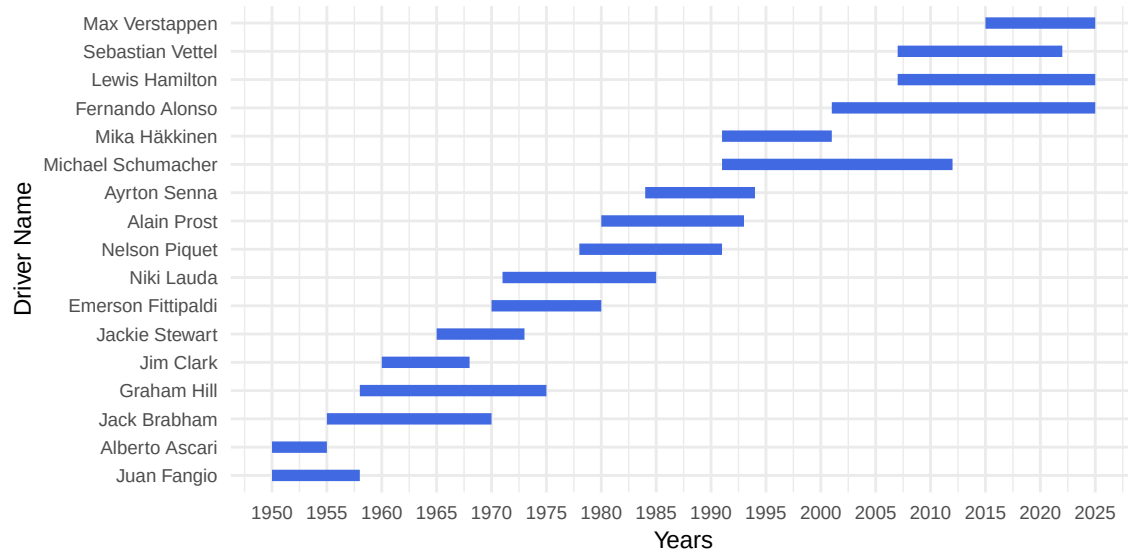


Figure 8: Careers

Some drivers enjoyed longer careers than others. For instance, Alonso, Schumacher, and Hamilton had the longest career spans, while Ascari, Clark, and Stewart had the shortest. The problem with this graph is that it does not effectively depict the dominance of any particular driver. For instance, Alonso secured his two titles in 2005 and 2006, and has continued racing since without winning any additional titles.

Now, if we replace the career duration with titles, where each segment represents the years between a driver's first and last title, we obtain the following graph :

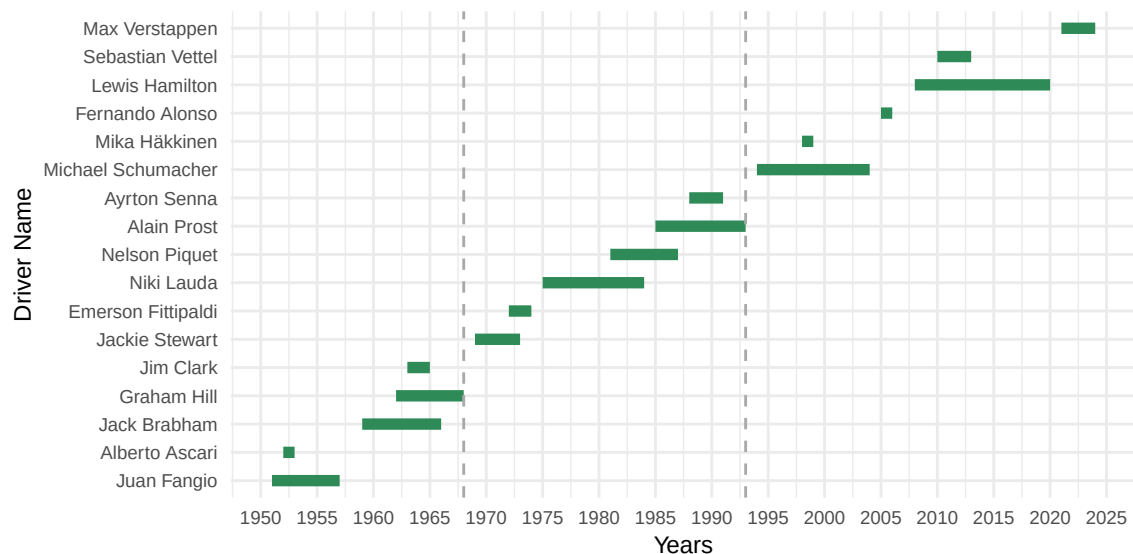


Figure 9: Titles

We divided this timeline into three phases: the first phase, the early stage, extends from the beginning until the late 1960s. The second phase begins with Jackie Stewart's first title in 1969 and continues until Michael Schumacher's first title in 1994. The final phase, or modern era, starts in 1994 and continues to the present.

In the plot above, each phase is marked with a vertical dashed grey line.

Now, when we display the career lengths based on the year of the first title, we can observe a gradual increase over the years :

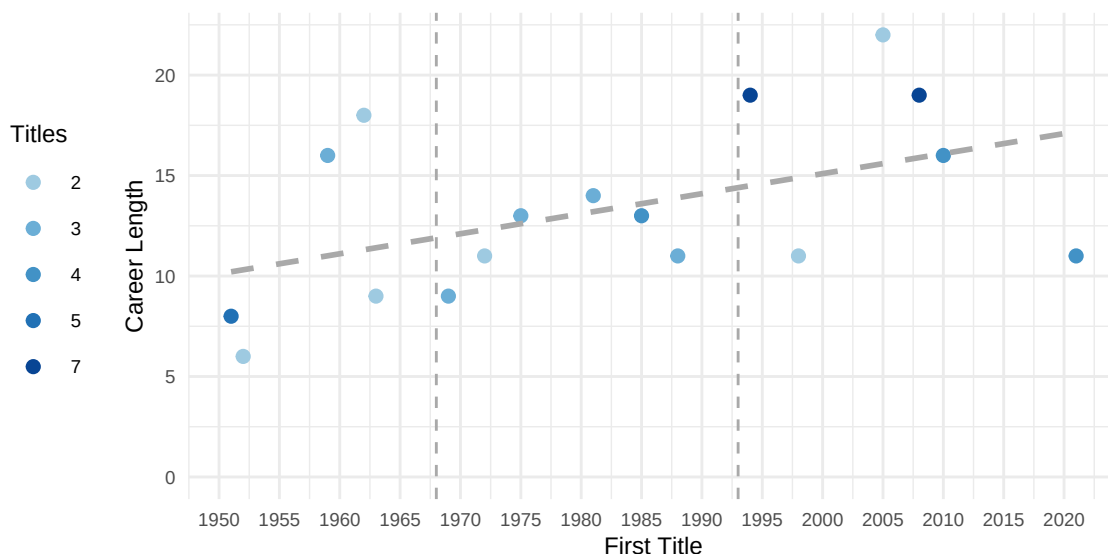


Figure 10: Career Start vs Career Length

A clear upward trend emerges, suggesting that drivers who started their careers in more recent decades tend to have longer careers. This may be attributed to various factors, including advancements in safety, improvements in car technology, and changes in race formats that allow for prolonged competition. Notable drivers in the modern era, such as Hamilton, benefit from these advancements, resulting in extended careers compared to drivers from earlier periods.

We can notice that while Juan Manuel Fangio was able to secure 5 titles within 8 years in the 1950s, Michael Schumacher and Lewis Hamilton achieved their 7 titles in 19 and 20 years respectively, with Hamilton still actively competing. This illustrates how the demands and dynamics of racing have shifted over time. The differentiation of data points by title count adds another layer of insight. Drivers with more titles, represented by larger and darker-colored points, typically have career lengths that align with the observed trend. The correlation suggests that those who excel and secure multiple championships also tend to enjoy longer careers, potentially due to their consistent performance, adaptability, and resilience in the competitive environment.

Conversely, while some historical drivers achieved significant success in fewer years, many of them faced different challenges, such as fewer races per season and a more hazardous racing environment. This context helps explain why earlier drivers, despite their talent, often have shorter career lengths represented in the plot.

The dashed line in the graph indicates the overall trend, reinforcing the notion that career length has increased over time for Formula 1 drivers. This trend not only reflects the effects of improved safety and technology but may also point to changes in how drivers manage their careers, including fitness and approach to racing.

4.1.3 Comparison

Now, let's compare the top drivers based on their wins and the number of races they have participated in :

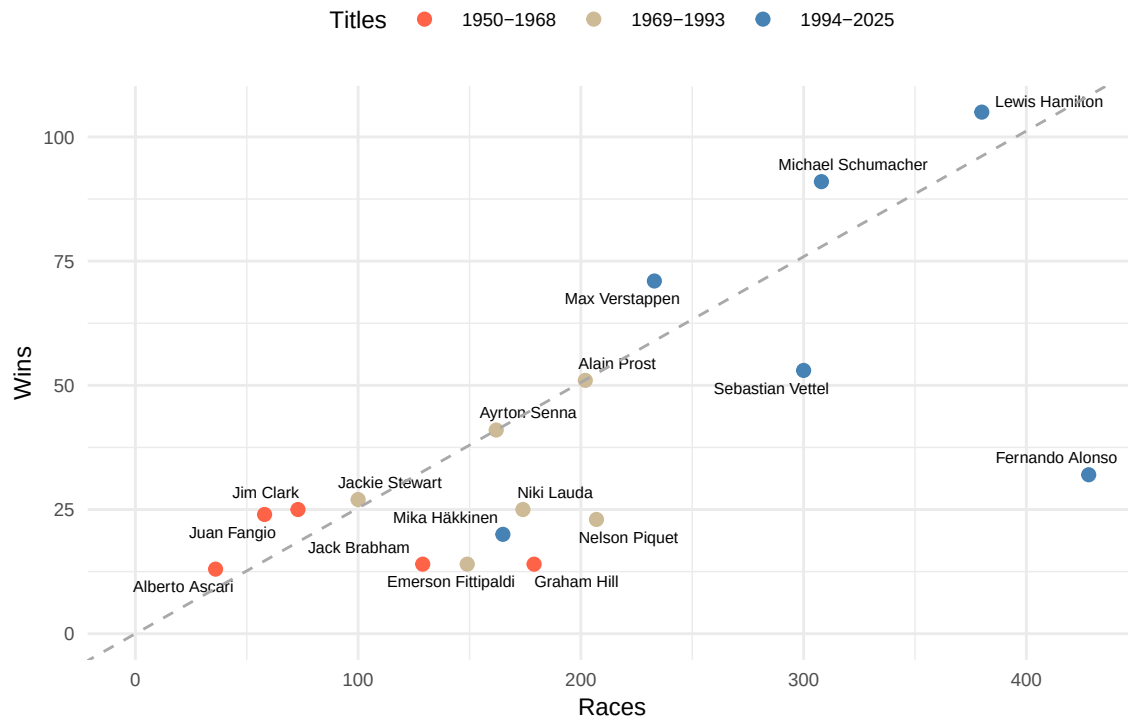


Figure 11: Wins vs Races

There is a noticeable upward trend in the plot, suggesting that drivers who participate in more races tend to achieve a higher number of wins. This correlation implies that experience and track time significantly contribute to success in the sport. Drivers from the 1994 to 2023 era, such as Hamilton and Vettel, are positioned in the top right quadrant, indicating both a high number of races and wins. The representation of this era in a distinct color underscores the advantages of modern technology and greater competitiveness, reflecting a notable shift in performance standards.

The varying colors used to group drivers by era emphasize the differences in victory rates and opportunities for success across time. Historical drivers may have achieved fewer wins but did so in a context with fewer races and different competitive landscapes. For instance, earlier eras, represented in warmer colors, often had fewer wins despite high title counts, pointing to the unique challenges faced by those drivers. Additionally, Fernando Alonso stands out for having a considerable number of races with only two titles, highlighting that a lengthy career does not guarantee championship success. This suggests that various external factors, such as team dynamics and competitive context, play significant roles in determining outcomes.

Overall, the plot illustrates that while participation in a greater number of races can lead to more wins, it does not exclusively dictate a driver's success. Factors like team performance, car technology, and the competitiveness of each era are crucial. The shifts in driver success across different decades indicate an evolution in Formula 1, influenced by changes in race formats and team strategies, as well as an expanding global audience. The color-coded groupings serve to highlight how drivers' experiences and achievements are shaped by the specific challenges and opportunities of their respective eras. This analysis provides a thorough understanding of the factors influencing driver success in Formula 1, as visualized through the plot, while also acknowledging the historical context that contributes to these differences.

And now the number of titles and seasons per driver :

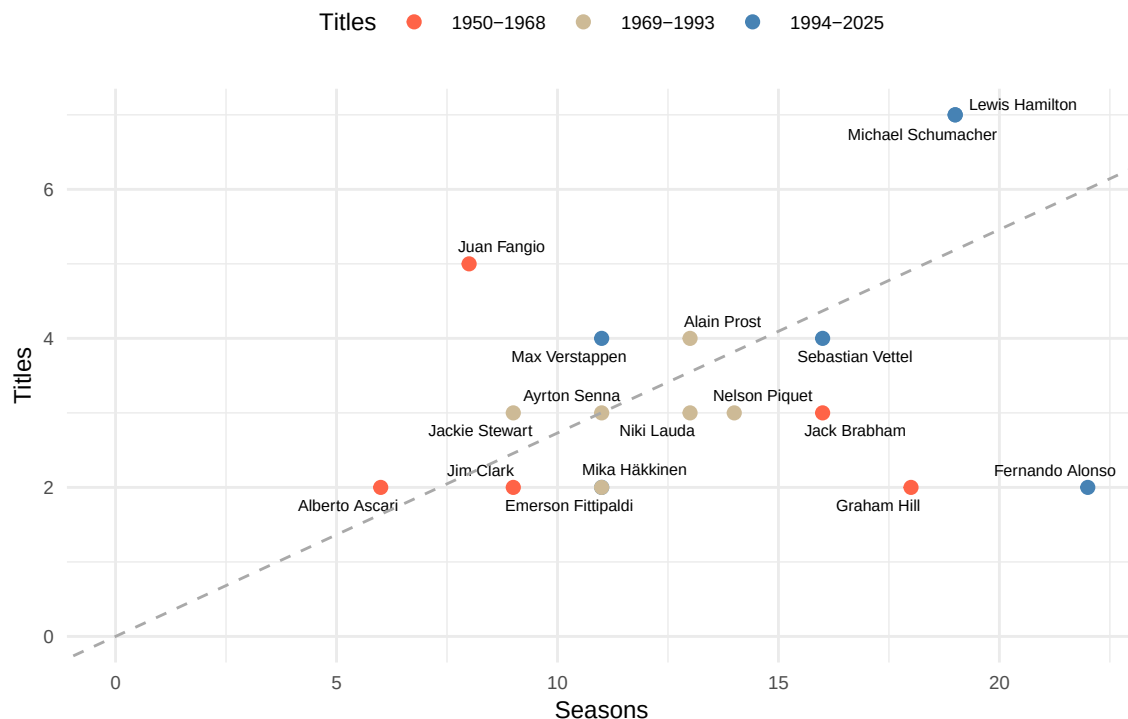


Figure 12: Titles vs Seasons

While the plot displays a trend where drivers with more seasons tend to win multiple titles, this is true for those who have achieved two or more titles. Notably, modern drivers such as Hamilton and Vettel, positioned in the top right quadrant, exemplify this trend, showcasing the advantage of extended careers combined with high performance. Their achievements in more recent years highlight the advancements in technology and team strategies that have influenced competitive outcomes.

The color coding of the data points emphasizes differences in performance across historical eras. For instance, drivers from the 1950 to 1988 timeframe, such as Fangio and Clark, achieved multiple titles within fewer seasons, reflecting the unique challenges of their competitive environments, which often had fewer races and different dynamics compared to today. This juxtaposition illustrates how the number of titles can vary significantly based on the era in which the drivers competed.

Additionally, the representation of drivers like Alonso showcases the relationship between seasons and titles, as he has spent a considerable number of seasons in the sport but has only two titles. This highlights that having a long career does not always correlate with championship victories, suggesting the impact of factors like team performance, adaptability, and the competitive landscape at any given time.

Overall, the plot underscores that while participation in more seasons can lead to multiple titles, it does not guarantee success. Various factors, including team dynamics, technological advancements, and the competitive environment in different eras, play crucial roles in determining a driver's achievements. The shifts in performance across different decades indicate an evolution within Formula 1, influenced by changing race formats and an increasingly global audience. This analysis provides a comprehensive understanding of how the number of seasons impacts title success while also recognizing the historical context that shapes these outcomes.

4.1.4 Summary

[We saw that recent drivers achieved better results with time, we also saw that some drivers were able to achieve better or same results in a shorter timeframe.

But this is not enough to determine if these drivers were really better than others. Would Fangio have achieved 10 titles have he run 16 seasons? What about the variation of number of races, points and rules over the seasons? And what about the car itself?]

We observed that recent drivers have generally achieved better results over time, indicating a trend of improved performance in modern Formula 1. This improvement can be attributed to a variety of factors, including advancements in technology, enhanced training practices, and improved team dynamics. However, it is equally noteworthy that some drivers from earlier eras managed to achieve comparable results within a significantly shorter timeframe. This raises an important question regarding the true measure of a driver's ability and whether career longevity directly equates to skill or success on the race track.

To determine if these recent drivers are genuinely better than their predecessors, we must consider multiple dimensions of comparison. For instance, would Juan Manuel Fangio have achieved 10 titles if he had raced for 16 seasons, similar to modern drivers? Fangio, who secured his five titles in just eight years during the 1950s, faced vastly different competitive landscapes and challenges than those in contemporary racing. His remarkable success in such a brief period prompts an investigation into context: how much of his success was due to his skill versus the circumstances and limitations of his era?

Additionally, the evolution of the sport itself complicates these comparisons. Over the years, Formula 1 has undergone significant changes in terms of race formats, scoring systems, and regulations. The number of races in a season has fluctuated, impacting a driver's opportunity to accumulate wins and points. In some seasons, drivers competed in fewer races, which naturally limits their chances for high totals, making it essential to contextualize statistics within the framework of their respective eras.

Points systems have also varied dramatically, altering how achievements are measured. Modern scoring rewards consistency and resilience across more races, which can skew comparisons against drivers from the past, who may have excelled under a different set of rules. For instance, Fangio's era awarded fewer points for victories, but achieving those wins was often subject to harsher competition and racing conditions.

Moreover, the cars themselves represent a crucial aspect of this discussion. The technology and engineering behind F1 cars have advanced exponentially over the decades, with modern vehicles benefiting from aerodynamics, data analysis, and materials that were not available in the past. This raises another question: how much of a driver's success is attributable to their talent and how much is due to the superior performance of the car? A skilled driver in a less competitive car might struggle when compared to a current driver operating one of today's high-tech machines, even if the former has a proven track record of wins.

Ultimately, comparing drivers across different eras poses a complex challenge. While it is tempting to label recent drivers as superior based on their successes, such determinations require a nuanced approach that considers a multitude of factors: career length, the evolution of racing rules, the technological trajectory of car design, and the competitive environment of their respective times. By acknowledging these dimensions, we gain a richer understanding of what it truly means to be "better" in the context of Formula 1, transcending mere statistics to appreciate the intricate tapestry of racing history and driver performance.

4.2 Championship

4.2.1 Races

The plot below illustrates the trend in the number of Formula 1 races held each year from 1950 to 2023. The x-axis represents the year, while the y-axis indicates the number of races. Each point reflects the actual number of races for that specific year, depicted in gray, and the variation among these points indicates fluctuations in race counts over the years, reflecting changes in the racing calendar and external factors affecting the sport.

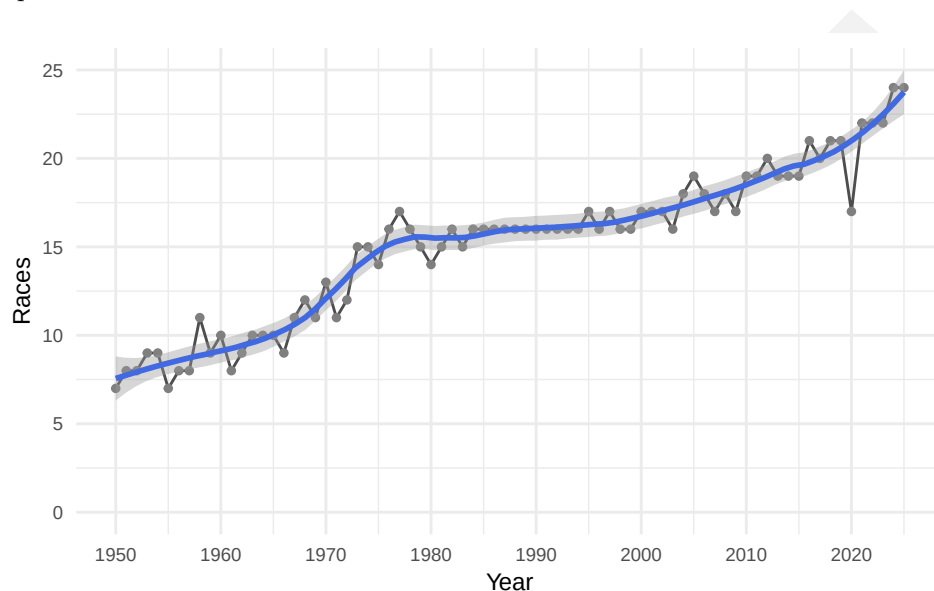


Figure 13: Races per Year

A clear upward trend is evident, starting from just 7 races in 1950 and gradually increasing, with several plateaus extending into the late 20th century. A more pronounced rise occurs in the 21st century, culminating in approximately 23 races by 2023, suggesting the growing popularity and expansion of Formula 1.

The loess smoothing curve, shown in royal blue, overlays the data points to provide a clearer representation of the overall trend in race numbers. This curve highlights the acceleration in race counts, particularly after 2000, indicating that Formula 1 has adapted to demand and increased the frequency of races. The shaded area around the smoothing line captures the confidence interval of the smoothed trend in a light blue hue, suggesting the level of uncertainty about the estimated values and demonstrating the variability in the data points over the years.

The plot conveys the evolution of Formula 1 through the number of races conducted annually. The sharp increase in races in later years reflects various factors, such as enhanced global interest in motorsports, changes in regulations, and the expansion of the racing calendar. The data's variability, particularly in the earlier decades, can be attributed to the formative years of the sport, which experienced significant changes in organization and promotion.

4.2.2 Points and Rules

The evolution of points systems and rules in Formula 1 has undergone significant changes over the years, reflecting the sport's adaptability to enhance competition and viewer engagement. Understanding these modifications is crucial for properly interpreting statistics and achievements across different eras of racing.

One of the most notable changes in the points system involves the method of calculating scores based on race results. In earlier years, drivers earned points for finishing in the top positions, but not all results were counted towards the championship. For instance, the introduction of the “**8 best results**” system allowed drivers to select their top performances to be counted toward their final tally. This change meant that a driver could have an off day in a few races but still remain competitive based on their standout performances. The revised approach promotes consistency and allows for some leeway, encouraging drivers to focus on achieving high results in selected races rather than simply accumulating points in every competition.

In recent years, the integration of new formats such as **sprint races** has further influenced the dynamics of scoring. Sprint events, which typically take place on weekends with traditional Grand Prix races, offer additional points opportunities. This development adds an exciting layer to the race weekend, allowing drivers to earn points not just through their main races but also by performing well in shorter sprints. The introduction of these sprints has reshaped strategies, pushing teams to adapt their approaches to maximize potential points across various race formats.

Another element that has gained attention is the inclusion of points awarded for the **fastest lap** during a race, further enriching the competitive landscape. By recognizing drivers who secure the fastest lap, this rule incentivizes aggressive racing strategies, compelling drivers to push their limits even when their overall position may not warrant it. This addition exemplifies how changes in rules can elevate the excitement and intensity of each race.

As we compare drivers and their results over time, it is important to note that we will not consider sprint races or fastest laps in our study. Our analysis will focus solely on race results, and we will take into account all points awarded across each driver's career. This approach allows us to maintain a more consistent basis for comparison, ensuring that our findings reflect the fundamental aspects of performance in Formula 1.

The implications of these transformed points systems extend beyond mere statistics. They affect team and driver strategies, influencing how races are approached. Teams may prioritize specific races or adopt riskier tactics, knowing that achieving the fastest lap or excelling in a sprint can contribute critically to their overall standings. This fosters a more vibrant competitive environment where drivers constantly seek opportunities to maximize their returns, complicating the notion of what it means to be successful in Formula 1.

Moreover, when comparing drivers across different eras, it becomes essential to contextualize their accomplishments within the frameworks established by the rules at the time. Historical figures like Juan Manuel Fangio, for example, achieved remarkable titles under a vastly different scoring system, contrasting sharply with the multifaceted approach seen today. Understanding these differences in rules and points systems allows fans and analysts to appreciate the evolution of the sport while acknowledging the unique challenges faced by drivers across generations.

The plot below shows the number of points given according to the race final position :

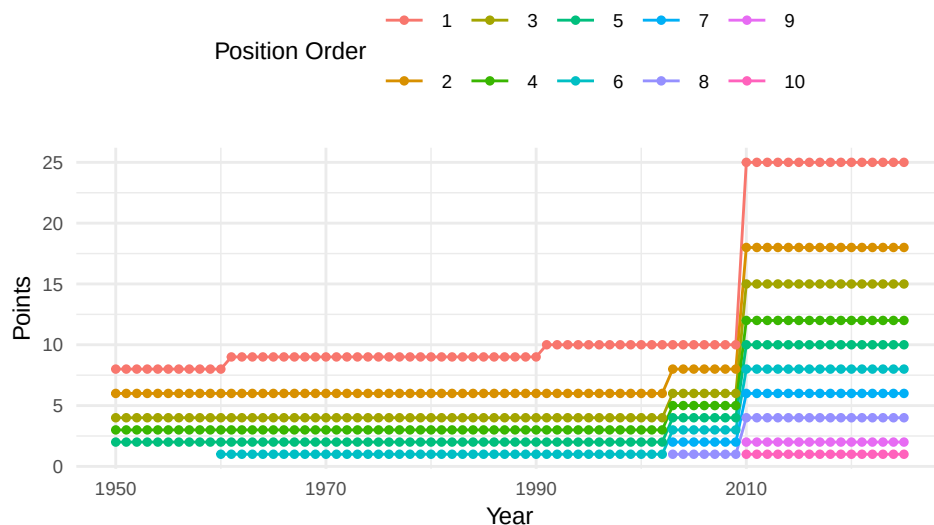


Figure 14: Points per Year

Until 1959, the scoring system for races awarded the winner 8 points, with an additional point granted for the fastest lap achieved during the race. However, in 1960, the point for the fastest lap was eliminated, and instead, the driver finishing in 6th position would receive one point. The following year, in 1961, the points awarded to the winner increased to 9, further evolving the scoring structure. By 1991, winners would receive 10 points, while other position points remained unchanged. The scoring system underwent a significant revision in 2003, when drivers other than the winner began to receive an additional 2 points. Additionally, drivers finishing in 7th and 8th positions would earn 2 points and 1 point, respectively. By 2010, the rules changed again, increasing the winner's points to 25 and extending points to the top 10 finishers.

Although not depicted in this analysis, it's worth noting that in 2019, a point for the fastest lap was reintroduced. Furthermore, between 1950 and 1990, only certain race results contributed towards the championship standings, indicating fluctuations in how scoring impacted overall competitive success during that period.

The plot below shows the percentage difference :

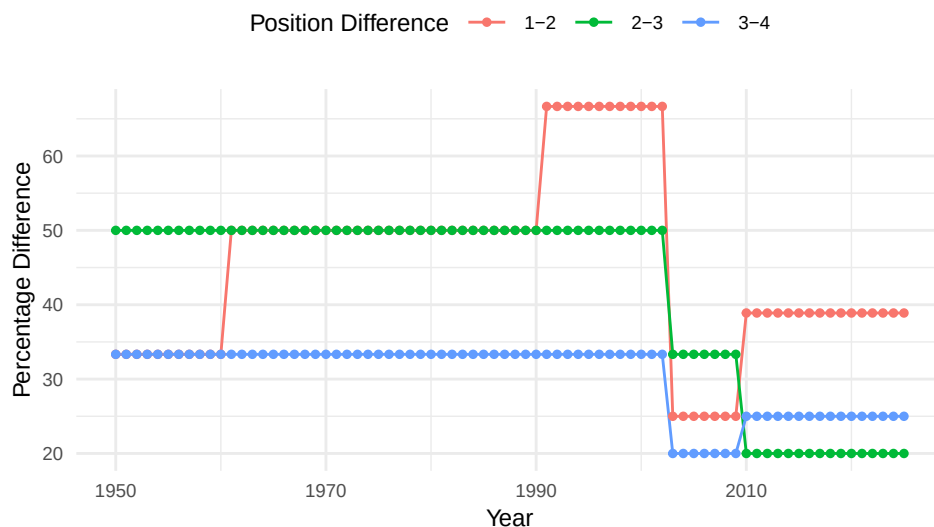


Figure 15: Percentage Difference

From 1950 until 1959, the winner received 33% more points than the second-place driver, who was awarded 50% more points than the driver finishing in third, while the third place earned 33% more points than the fourth. However, in 1960, the rules shifted so that the winner received 50% more points than the second-place driver, and this percentage increased to 66% in 1991. Starting in 2003, this difference in points began to decrease significantly, with a slight increase in 2010, resulting in the winner obtaining approximately 40% more points than the second-place finisher, who in turn received 20% more than the third, while the third place earned 25% more than the fourth.

This plot illustrates how the importance of finishing position has diminished since 2003, suggesting that the risks associated with overtaking another driver may no longer justify the potential rewards. As drivers weigh the dangers of overtaking, the incentive structure of points allocation reflects a changing competitive landscape, where maintaining position may become a more strategic choice.

The table below displays championships in which the driver who earned the highest total points did not win the title. This occurred because only a portion of these points were counted, allowing the second-place driver to secure the championship with a greater proportion of their retained points :

year	driverNameChampion	pointsChampion	driverNameBest	pointsBest
1964	 John Surtees	40(40)	 Graham Hill	41(39)
1988	 Ayrton Senna	94(90)	 Alain Prost	105(87)

In the world of Formula 1 racing, championship titles are not solely determined by the total points a driver accumulates over the season. This nuance became particularly evident in dramatic cases such as those of John Surtees in 1964 and Ayrton Senna in 1988.

In 1964, John Surtees clinched the World Championship with a total of 40 points, and importantly, all of these points were eligible for inclusion in the championship standings. Surtees faced stiff competition from his rival, Graham Hill, who ended the season with 41 points—just a single point ahead. However, not all of Hill's points were counted towards the championship, as only 39 points were eligible, thus causing him to lose the title to Surtees. This incident serves as a poignant reminder of the impact of point retention rules in determining the final standings, which can overshadow pure performance metrics.

Fast forward to 1988, when the same theme of point retention played out yet again in a fierce rivalry between Ayrton Senna and Alain Prost. Senna finished the season with an impressive 94 points, out of which 90 points were counted towards his championship total. Conversely, Alain Prost amassed a remarkable 105 points but faced a significant setback as only 87 of these points were deemed eligible for the championship classification. This discrepancy ultimately placed Prost in second position, despite his overall higher point total.

The plot below shows for every world champion, the percentage of wins over the season :

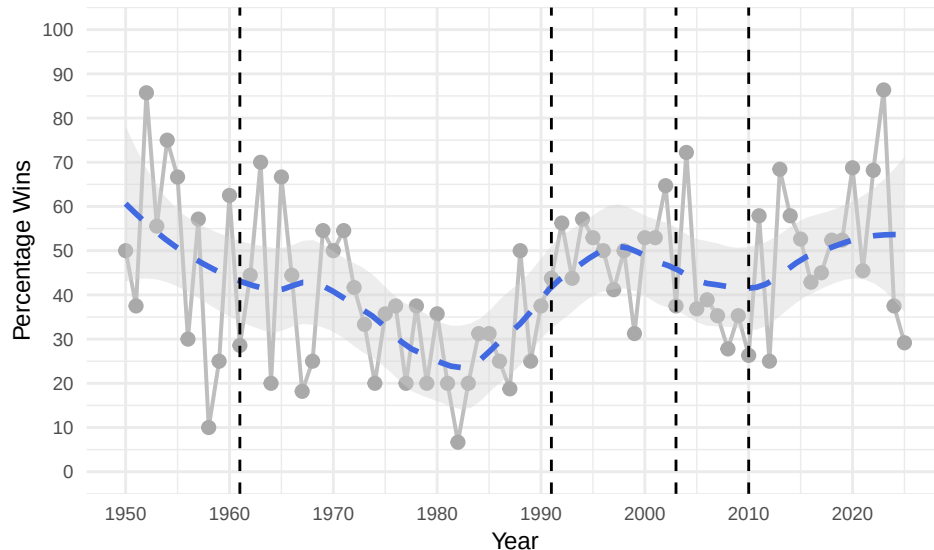


Figure 16: Percentage Wins

We can see that from 1965 and Jim Clark's title, until 2004 and Michael Schumacher third or fourth title, that is a period of 40 years, no pilot achieved a win to race ratio over 65%. And if we filter out the earlier championships with less than 12 races (where a high percentage is easier to achieve than when there are more races), the only high percentage championships are the recent ones, that is after 2000.

The two highest percentages are when Max Verstappen won the title in 2023 with 19 races won over 22, and Alberto Ascari with 6 wins over 7 races. Both achieved more than 85% win success.

year	driverName	nbraces	wins	winspc
2023	 Max Verstappen	22	19	86.4
1952	 Alberto Ascari	7	6	85.7
1954	 Juan Fangio	8	6	75.0
2004	 Michael Schumacher	18	13	72.2
1963	 Jim Clark	10	7	70.0
2020	 Lewis Hamilton	16	11	68.8
2013	 Sebastian Vettel	19	13	68.4
2022	 Max Verstappen	22	15	68.2
1955	 Juan Fangio	6	4	66.7
1965	 Jim Clark	9	6	66.7

On the other side, it is possible to win the championship by winning less races. The table below shows the drivers who won the championship with the lowest win to race ratio :

year	driverName	nbraces	wins	winspc
1964	 John Surtees	10	2	20.0
1974	 Emerson Fittipaldi	15	3	20.0
1977	 Niki Lauda	15	3	20.0
1979	 Jody Scheckter	15	3	20.0
1981	 Nelson Piquet	15	3	20.0
1983	 Nelson Piquet	15	3	20.0
1987	 Nelson Piquet	16	3	18.8
1967	 Denny Hulme	11	2	18.2
1958	 Mike Hawthorn	10	1	10.0
1982	 Keke Rosberg	15	1	6.7

We can see that from 1974 until 1987, it was very common to win the world title with less than 20% races won. Notably, the Brazilian driver Nelson Piquet won his 3 world titles by winning less than 20% of the races each time.

The record being held by Keke Rosberg who won the 1982 championship by winning only 1 race out of 15.

Note that in 2025 the win to races ratio was one of the lowest in the recent years : with only 29% wins, Lando Norris was able to win the title, even with less victories (7) than Verstappen (8). His teammate Piastri was able to win the same number of races.

The plot below shows the total number of points given per year :

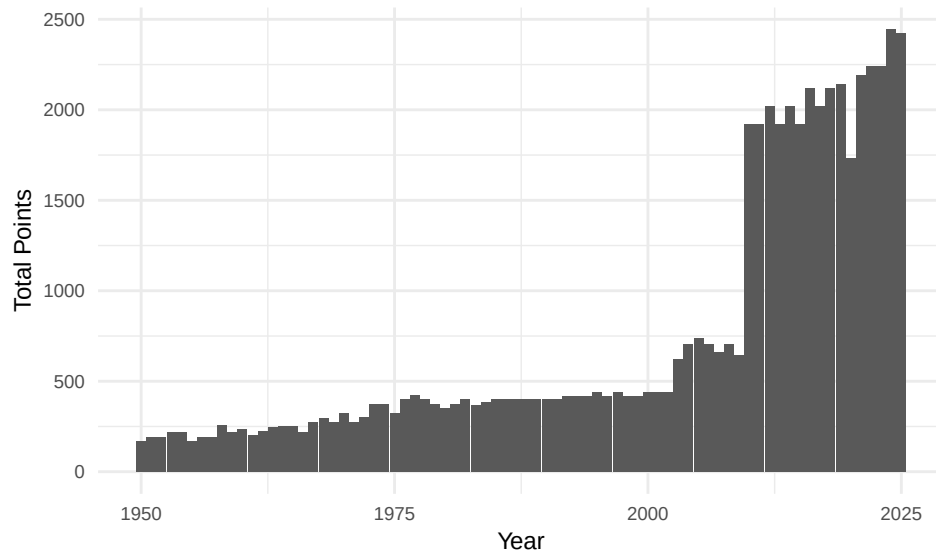


Figure 17: Total Points per Year

In conclusion, the changes in points and rules over time significantly impact how success and performance are measured in Formula 1. The transition from early systems to contemporary practices reflects the sport's commitment to enhancing competitiveness and excitement. Analyzing these developments provides crucial context for assessing driver achievements and understanding the strategic decisions made by teams and drivers alike. The interplay between points systems, race formats, and performance metrics illustrates the complexity of the sport and highlights how Formula 1 continually evolves to captivate its audience.

4.3 Constructors

Constructor's results over time

DRAFT

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